

Car Plate Recognition and Reconstruction with Deep Learning

Computer Vision – A.Y. 2024-2025

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Outline

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Datasets and metrics

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Introduction

Our goal was to develop a computer vision system to **detect license plates in images** using bounding box localization and extract text through optical character recognition, handling varied lighting and viewing angles for real-world accuracy.

1

Baseline Development

Create proprietary baseline model

2

Model Implementation

Develop Yolov5 and PDLPR models

3

Testing & Evaluation

Benchmark all models

4

Performance Comparison

Compare results and document findings

Yolov5-PDLPR Architecture Overview



Object Detection:
Yolov5

PDLPR

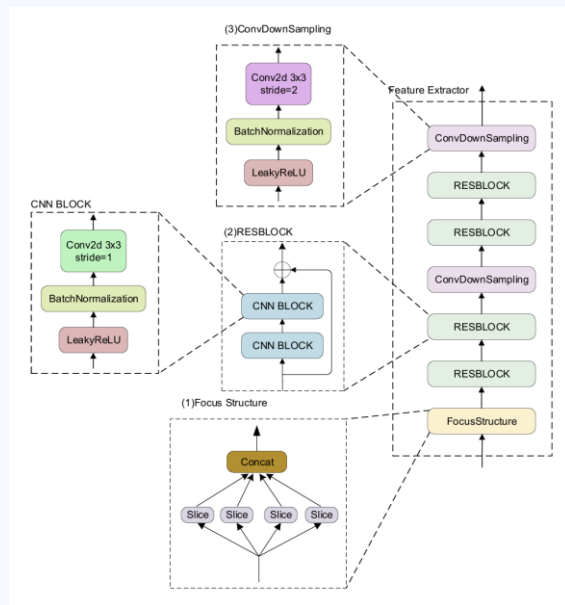
IGFE

Encoder

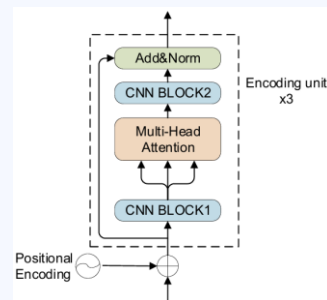
Parallel
Decoder

皖AR0011

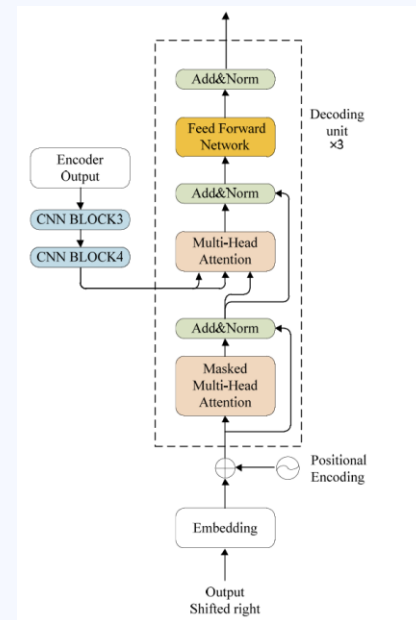
How PDLPR works



IGFE



Encoder



Decoder

ResNet18 Architecture Overview



Same
Convolutional
Architecture:
ResNet18

1

**Bounding
Box**
linear part



2

**OCR linear
part**

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CCPD (Chinese City Parking Dataset)

A large-scale, publicly available dataset for license plate detection and recognition. Contains over **250,000** images of vehicles with annotated license plates.

Dataset Characteristics:

- Diverse conditions: various lighting, weather, and viewpoints.
- Plates captured in real-world scenarios including occlusion, blur, and distortion.
- Annotations include bounding boxes for plates and character-level labels.



Examples of possible car plates

fn



db



tilt



rotate



blur



weather



The Experimental setup (1)

Yolov5

- **Loss function:** GloU loss
- **Optimizer:** Adam, with LR: 0.001
- **Batch:** 50
- **Epochs:** 15

PDLPR

- **Loss function:** Cross Entropy
- **Optimizer:** Adam, with LR: $1e^{-4}$
- **Batch:** 128
- **Epochs:** 50

$$L_{IoU} = 1 - IoU(B_{Pred}, B_{gt})$$
$$IoU(A, B) = \frac{A \cap B}{A \cup B}$$

$$L_{OCR} = -\frac{1}{T} \sum_{i=1}^T \sum_{k=1}^C y_{i,k} \log(\hat{y}_{i,k})$$

The Experimental setup (2)

RN18 Bounding Box

- **Loss function:** Mean Squared Error
- **Optimizer:** Adam, with LR: $1e^{-3}$
- **Batch:** 50
- **Epochs:** 30

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

RN18 OCR

- **Loss function:** Cross Entropy
- **Optimizer:** Adam, with LR: $1e^{-4}$, weight decay: $5e^{-5}$
- **Batch:** 128
- **Epochs:** 30

$$L_{OCR} = -\frac{1}{T} \sum_{i=1}^T \sum_{k=1}^C y_{i,k} \log(\hat{y}_{i,k})$$

Yolov5 and RN18 BB results

IoU > 0.70

dataset	Yolov5	RN18 OCR
base	96,8%	88.7%
blur	89,4%	80.1%
challenge	87,7%	82.1%
db	78,3%	76.2%
fn	78,4%	75.8%
rotate	89,2%	84.3%
tilt	87,2%	82.4%
weather	92,5%	86.2%

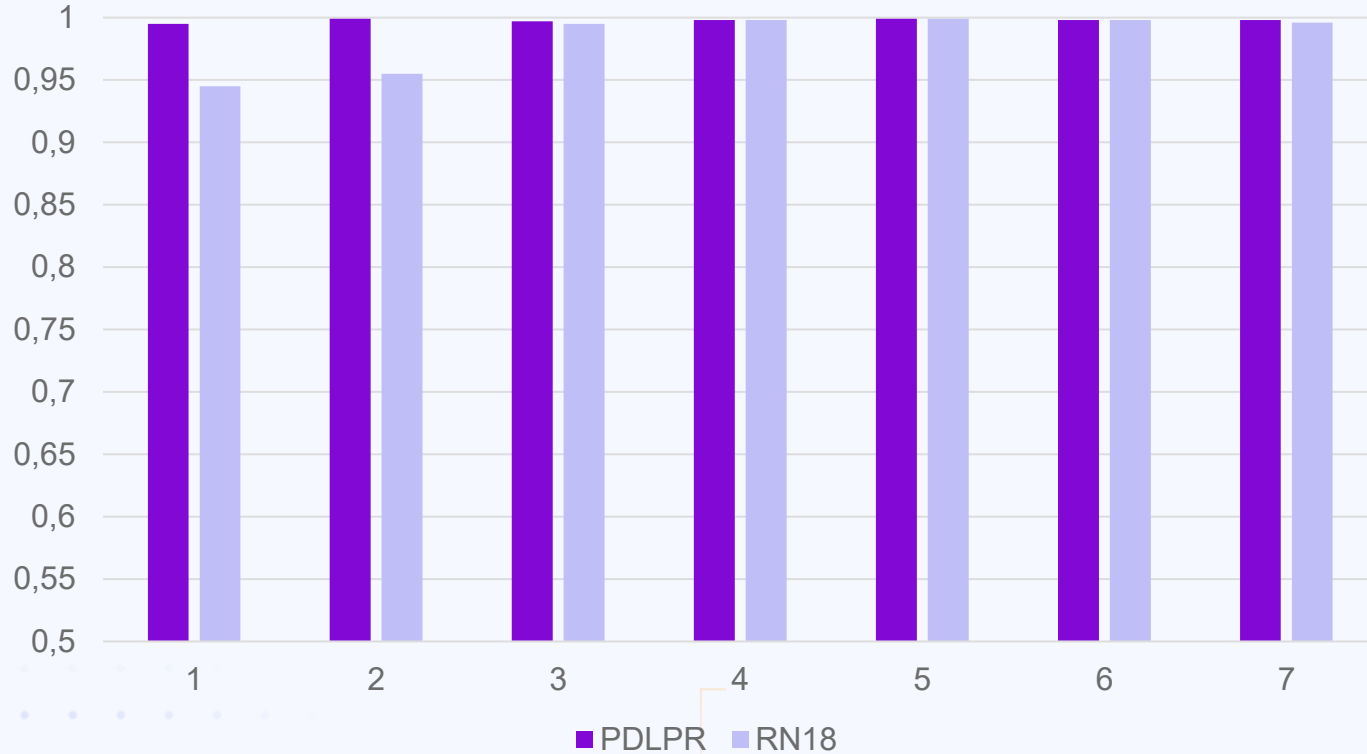
RN18 OCR results

	base	tilt	blur	rotate	challenge	fn	db	weather	total
samples	40000	30216	20611	10053	50003	20967	10132	9999	150K
avg loss	0.3216	0.1204	0.1077	0.0700	0.0734	0.0859	0.1089	0.0040	0.1115
acc/char	99.45%	88.65%	95.44%	96.14%	94.64%	93.94%	93.28%	98.60%	95.14%
acc/plate	98.86%	75.39%	79.45%	84.12%	78.49%	77.51%	76.41%	97.40%	83.03%

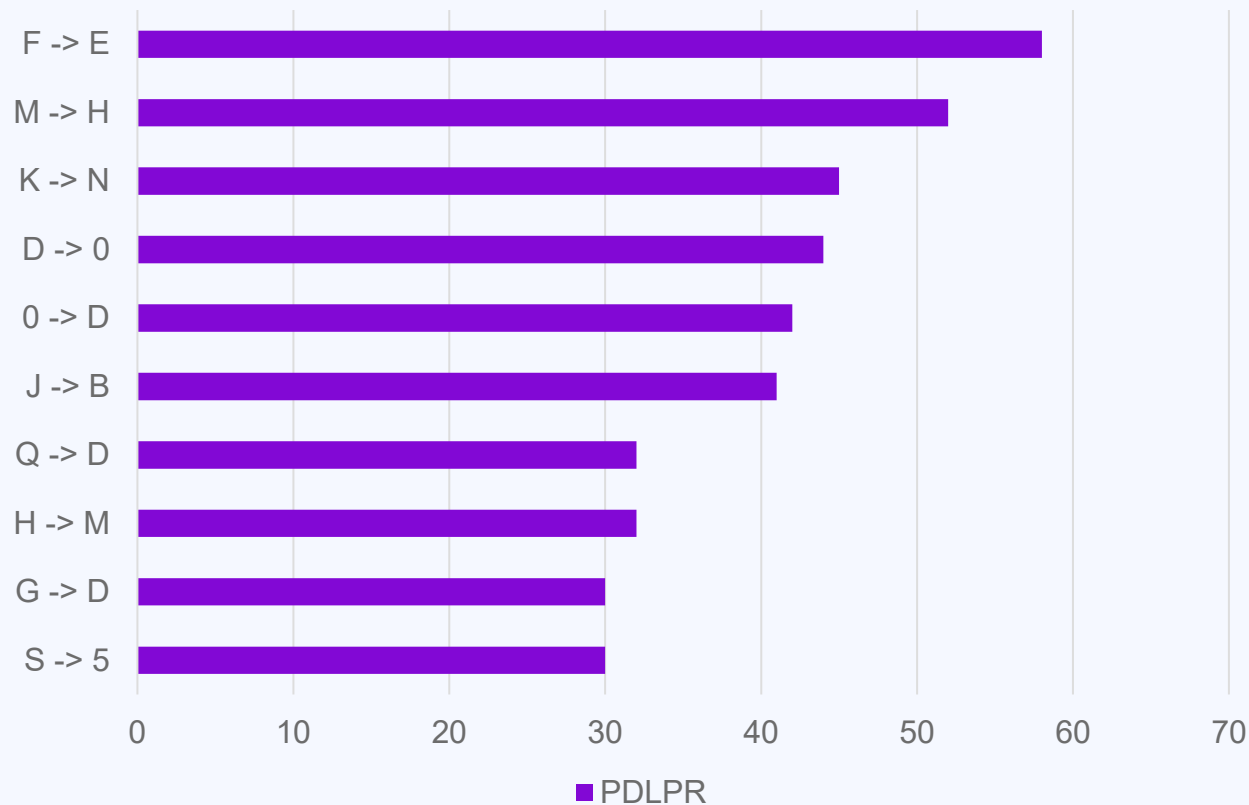
PDLPR results

	base	tilt	blur	rotate	challenge	fn	db	weather	total
samples	40000	30216	20611	10053	50003	20967	10132	9999	150K
avg loss	0.1096	0.0922	0.0552	0.1178	0.1206	0.1511	0.0080	0.1096	0.1115
acc/char	99.45%	96.74%	97.44%	98.74%	97.11%	96.82%	96.02%	99.78%	97.76%
acc/plate	99.07%	82.22%	85.83%	92.86%	85.94%	84.27%	80.04%	98.46%	89.59%

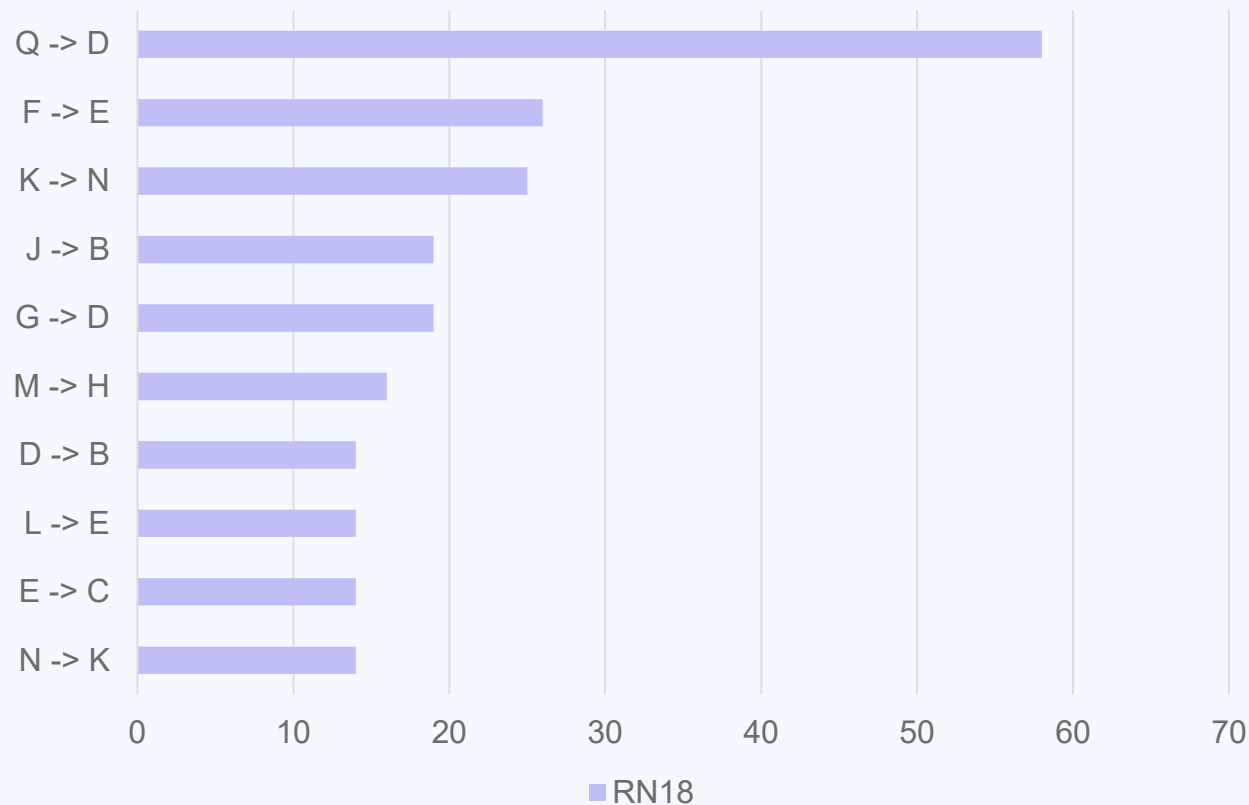
Accuracy for character position (ccpd-weather)



Most common letter and number confusions (1)



Most common letter and number confusions (2)





Conclusions

The specialized **YOLOv5 + PDLPR pipeline** delivers superior results, while **ResNet18**, a general-purpose model, achieves comparable performance with significantly faster training and easier management.

ResNet18 OCR and PDLPR show **comparable performance**, as ResNet18 demonstrates superior character recognition capabilities. However, ResNet18 falls slightly behind in bounding box detection, though it still delivers absolutely acceptable results that meet project requirements.

Both approaches effectively deliver the desired outcomes while offering distinct advantages in specialization versus versatility.



Future works

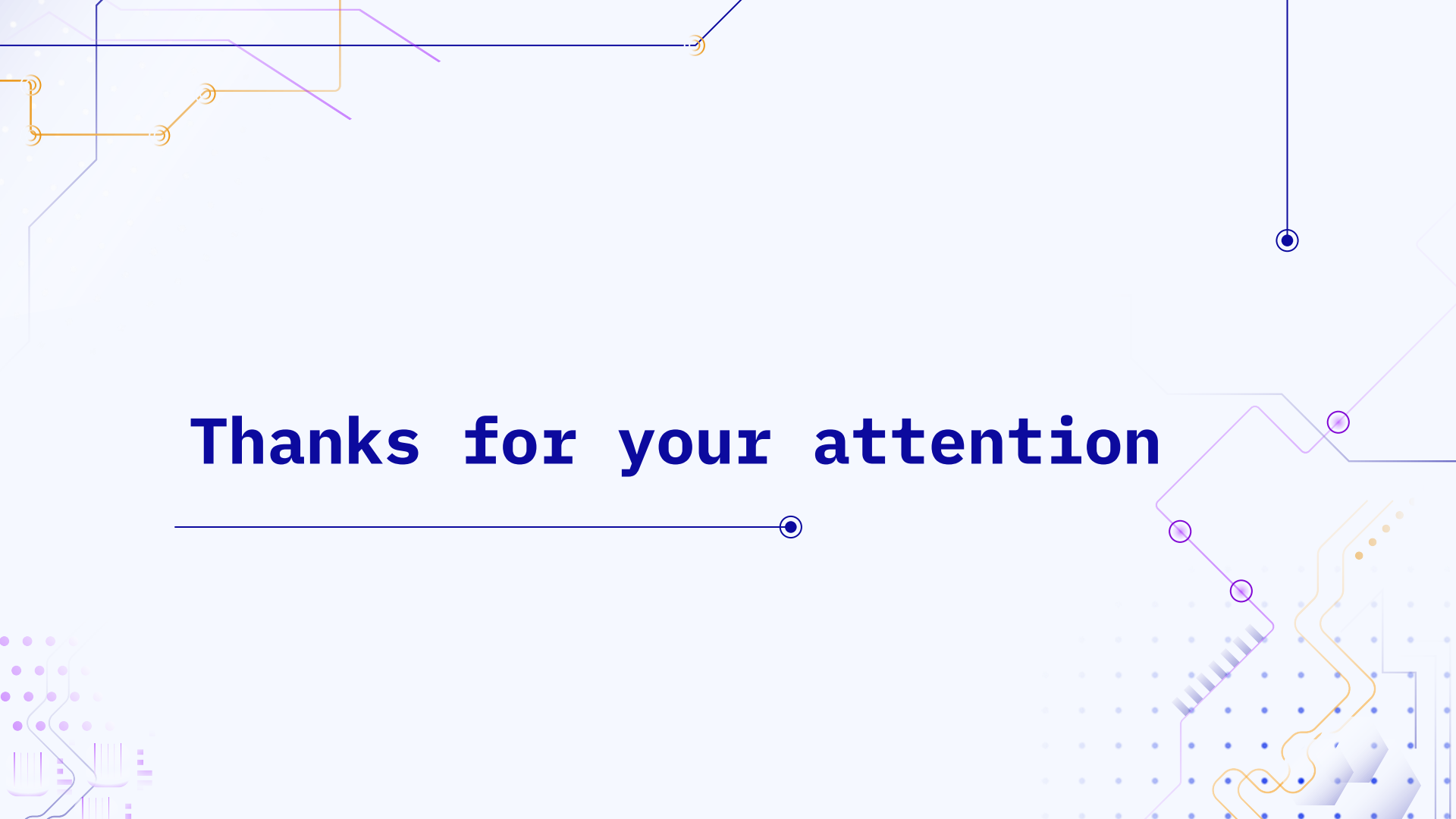
It would be interesting to evaluate the model proposed in the paper using **YOLOv11**, the latest version of the YOLO architecture, to potentially improve detection performance.

Testing the model on **additional datasets** would help assess its **flexibility** and **generalization capabilities**, especially for different types of license plates across various regions or formats.



Reference

1. Tao, L., A Real-Time License Plate Detection and Recognition Model in Unconstrained Scenarios. Sensors, Hong, S., Lin, Y., Chen, Y., He, P. and Tie, Z. (2024). 24(9), 2791
2. Xu, Z.; Yang, W.; Meng, A.; Lu, N.; Huang, H.; Ying, C.; Huang, L. Towards end-to-end license plate detection and recognition: A large dataset and baseline. In Proceedings of the European Conference on Computer Vision (ECCV), Munich, Germany, 8–14 September 2018.
3. R. K. Prajapati, Y. Bhardwaj, R. K. Jain and D. Kamal Kant Hiran, "A Review Paper on Automatic Number Plate Recognition using Machine Learning : An In-Depth Analysis of Machine Learning Techniques in Automatic Number Plate Recognition: Opportunities and Limitations," 2023 International Conference on Computational Intelligence, Communication Technology and Networking (CICTN), Ghaziabad, India, 2023, pp. 527–532



Thanks for your attention